

# Uber Ride Analysis

*Transforming Ride Data into Actionable Business  
Insights*

*A Dual-Dashboard Analysis Built Using Tableau*

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**ROLE : DATA ANALYST**

**TOOLS USED :**

- EXCEL,
- TABLEAU DESKTOP.

**DASHBOARD 1:**

**BUSINESS PERFORMANCE OVERVIEW**

**DASHBOARD2 :**

**RIDE & CUSTOMER ANALYSIS**

## **ABSTRACT**

This project focuses on analyzing Uber ride booking data to understand overall business performance and customer behavior. The raw dataset was first cleaned and prepared using Excel, after which it was imported into Tableau for visualization. Two interactive dashboards were created: one to monitor high-level business KPIs such as trips and revenue, and another to analyze ride patterns, cancellations, ratings, and operational efficiency. The insights derived from these dashboards support data-driven decision-making for improving ride operations and customer experience.

## **OVERVIEW**

Ride-hailing platforms like Uber generate large volumes of transactional and behavioral data. Analyzing this data is critical to understanding demand trends, customer satisfaction, and operational efficiency. This project provides a structured Tableau-based analysis that converts raw ride data into meaningful insights through KPIs, standard charts, and advanced visualizations.

## **PROBLEM STATEMENT**

Ride-hailing companies often face challenges in identifying demand patterns, peak hours, cancellation behavior, and service quality issues. Without proper visualization and analysis, it becomes difficult to optimize operations and improve customer satisfaction. This project aims to analyze Uber ride data to uncover these patterns and provide actionable insights using Tableau dashboards.

## **DATA SOURCE & DATA COLLECTION**

The project uses a cleaned Uber ride booking dataset in Excel/CSV format. The dataset contains ride-level transactional details including booking information, location data, fare details, ride status, and customer feedback metrics.

## **DATA FIELDS USED**

- Booking Id
- Trip Date
- Pickup City
- Drop City
- Vehicle Type
- Booking Status
- Fare Amount / Booking Value
- Payment Method
- Customer Rating
- CTAT / VTAT

## **TOOLS USED**

- ✓ Microsoft excel
- ✓ Tableau desktop

|    | A          | B        | C             | D                     | E             | F             | G                   | H                 | I        | J        | K             | L              | M               |
|----|------------|----------|---------------|-----------------------|---------------|---------------|---------------------|-------------------|----------|----------|---------------|----------------|-----------------|
|    | date       | time     | booking_id    | booking_status        | customer_id   | vehicle_type  | pickup_location     | drop_location     | avg_vtat | avg_ctst | booking_value | payment_method | customer_rating |
| 1  | 23-3-2024  | 12:29:38 | "CNR5884300"  | No Driver Found       | "C01982111"   | eBike         | Palam Vihar         | Jhilmil           | 8        | 29       | 414           | Unknown        | 4.5             |
| 2  | 29-11-2024 | 18:01:39 | "CNR1326809"  | Incomplete            | "C014604802"  | Go Sedan      | Shastri Nagar       | Gurgaon Sector 56 | 5        | 14       | 237           | UPI            | 4.5             |
| 3  | 23-8-2024  | 08:56:10 | "CNR8494506"  | Completed             | "C019202816"  | Auto          | Khandsa             | Malviya Nagar     | 13       | 26       | 627           | Debit Card     | 4.9             |
| 4  | 21-10-2024 | 17:17:25 | "CNR8906825"  | Completed             | "C012610914"  | Premier Sedan | Central Secretariat | Inderlok          | 13       | 29       | 416           | UPI            | 5.0             |
| 5  | 16-9-2024  | 22:08:00 | "CNR1950162"  | Completed             | "C019933542"  | Bike          | Ghitorni Village    | Khan Market       | 5        | 20       | 737           | UPI            | 4.3             |
| 6  | 6-2-2024   | 09:44:56 | "CNR4096693"  | Completed             | "C014670564"  | Auto          | AIMS                | Narsinghpur       | 5        | 18       | 316           | UPI            | 4.6             |
| 7  | 17-6-2024  | 15:45:58 | "CNR2002539"  | Completed             | "C016800553"  | Go Mini       | Vaishali            | Punjabi Bagh      | 7        | 20       | 640           | UPI            | 4.1             |
| 8  | 19-3-2024  | 17:37:37 | "CNR6568000"  | Completed             | "C018610436"  | Auto          | Mayur Vihar         | Cyber Hub         | 12       | 17       | 136           | UPI            | 4.2             |
| 9  | 14-9-2024  | 12:49:09 | "CNR75110807" | No Driver Found       | "C017873618"  | Go Sedan      | Noida Sector 62     | Noida Sector 18   | 8        | 29       | 414           | Unknown        | 4.5             |
| 10 | 16-12-2024 | 19:06:48 | "CNR7721892"  | Incomplete            | "C015214275"  | Auto          | Rohini              | Adarsh Nagar      | 6        | 26       | 135           | Cash           | 4.5             |
| 11 | 14-6-2024  | 16:24:12 | "CNR9070334"  | Completed             | "C016680340"  | Auto          | Udyog Bhawan        | Dwarka Sector 21  | 8        | 19       | 181           | Cash           | 4.9             |
| 12 | 18-9-2024  | 08:09:38 | "CNR9519127"  | No Driver Found       | "C017568143"  | Auto          | Vidhan Sabha        | AIMS              | 8        | 29       | 414           | Unknown        | 4.5             |
| 13 | 25-6-2024  | 22:44:15 | "CNR4386945"  | Cancelled by Driver   | "C015543520"  | eBike         | Patel Chowk         | Kherki Daula Toll | 5        | 29       | 414           | Unknown        | 4.5             |
| 14 | 11-9-2024  | 19:29:39 | "CNR2987763"  | Completed             | "C012669710"  | Go Mini       | Malviya Nagar       | Ghitorni Village  | 12       | 28       | 394           | UPI            | 4.7             |
| 15 | 18-10-2024 | 18:28:53 | "CNR8962232"  | Completed             | "C01789354"   | Go Mini       | Madipur             | GTB Nagar         | 14       | 31       | 836           | UPI            | 4.4             |
| 16 | 7-6-2024   | 15:05:35 | "CNR2390352"  | Completed             | "C015432215"  | Auto          | Jama Masjid         | Khan Market       | 9        | 37       | 410           | Uber Wallet    | 4.9             |
| 17 | 1-7-2024   | 10:51:16 | "CNR3221338"  | Completed             | "C012581698"  | Premier Sedan | IGI Airport         | Madipur           | 6        | 28       | 401           | UPI            | 4.3             |
| 18 | 15-12-2024 | 15:08:25 | "CNR6739317"  | Cancelled by Driver   | "C018682675"  | Go Sedan      | Vinobapuri          | GTB Nagar         | 6        | 29       | 414           | Unknown        | 4.5             |
| 19 | 24-11-2024 | 09:07:10 | "CNR6126048"  | Cancelled by Customer | "C011060329"  | eBike         | Kashmere Gate       | Anand Vihar       | 12       | 29       | 414           | Unknown        | 4.5             |
| 20 | 24-5-2024  | 19:53:57 | "CNR9465840"  | Cancelled by Driver   | "C019046501"  | eBike         | Pitampura           | Rajiv Nagar       | 10       | 29       | 414           | Unknown        | 4.5             |
| 21 | 14-12-2024 | 21:27:04 | "CNR843229"   | Completed             | "C014929303"  | Go Sedan      | Punjabi Bagh        | Mansarovar Park   | 6        | 34       | 264           | Uber Wallet    | 4.7             |
| 22 | 29-7-2024  | 06:28:22 | "CNR7911434"  | Completed             | "C018465004"  | Auto          | Greater Noida       | Botanical Garden  | 7        | 21       | 183           | UPI            | 4.2             |
| 23 | 23-2-2024  | 11:02:45 | "CNR6178330"  | Completed             | "C017719678"  | Go Mini       | Tis Hazari          | IMT Manesar       | 10       | 25       | 713           | UPI            | 3.6             |
| 24 | 9-3-2024   | 20:14:24 | "CNR5309981"  | Completed             | "C013870048"  | Go Mini       | Noida Sector 18     | Old Gurgaon       | 8        | 41       | 113           | UPI            | 4.9             |
| 25 | 28-8-2024  | 16:30:14 | "CNR2978950"  | Completed             | "C0182978950" | Auto          | Kanhaiya Nagar      | Anand Vihar       | 14       | 25       | 767           | Debit Card     | 4.6             |
| 26 | 8-8-2024   | 06:34:38 | "CNR2806943"  | Completed             | "C017550481"  | Uber XL       | Okhla               | Barakhamba Road   | 5        | 21       | 495           | Credit Card    | 4.2             |
| 27 | 26-1-2024  | 09:44:00 | "CNR83614535" | Cancelled by Driver   | "C012507102"  | Auto          | Cyber Hub           | Saket             | 12       | 29       | 414           | Unknown        | 4.5             |
| 28 | 12-4-2024  | 19:42:35 | "CNR4499383"  | No Driver Found       | "C015717521"  | Premier Sedan | Sadar Bazar Gurgaon | Mehrauli          | 8        | 29       | 414           | Unknown        | 4.5             |
| 29 | 5-4-2024   | 18:57:12 | "CNR834201"   | Incomplete            | "C017891331"  | Bike          | Shastri Park        | Botanical Garden  | 3        | 28       | 304           | UPI            | 4.5             |
| 30 | 3-11-2024  | 12:24:59 | "CNR8779845"  | Completed             | "C017444351"  | Auto          | Faridabad Sector 15 | Malviya Nagar     | 9        | 31       | 2847          | Debit Card     | 4.6             |
| 31 | 18-5-2024  | 17:37:52 | "CNR2178654"  | Cancelled by Driver   | "C013631860"  | Go Mini       | Qutub Minar         | Vishwavidyalaya   | 9        | 29       | 414           | Unknown        | 4.5             |
| 32 | 15-11-2024 | 05:53:55 | "CNR3871541"  | Completed             | "C019849259"  | Go Mini       | Mundka              | Preet Vihar       | 9        | 41       | 282           | UPI            | 4.4             |
| 33 | 4-2-2024   | 18:08:04 | "CNR2877692"  | Completed             | "C015901711"  | Premier Sedan | DLF City Court      | Nehru Place       | 12       | 21       | 656           | Uber Wallet    | 4.9             |
| 34 | 29-5-2024  | 14:05:10 | "CNR9603633"  | Completed             | "C015856503"  | Auto          | Tis Hazari          | Shahdara          | 2        | 19       | 134           | Credit Card    | 4.4             |
| 35 | 13-7-2024  | 19:53:15 | "CNR8087051"  | Cancelled by Driver   | "C019716528"  | Go Sedan      | New Colony          | Ghitorni Village  | 5        | 29       | 414           | Unknown        | 4.5             |

|    | A          | B        | C             | D                     | E            | F             | G                         | H                         | I  | J  | K    | L           | M   |
|----|------------|----------|---------------|-----------------------|--------------|---------------|---------------------------|---------------------------|----|----|------|-------------|-----|
| 36 | 13-7-2024  | 19:53:15 | "CNR8087051"  | Cancelled by Driver   | "C019716528" | Go Sedan      | New Colony                | Ghitorni Village          | 5  | 29 | 414  | Unknown     | 4.5 |
| 37 | 7-6-2024   | 19:21:10 | "CNR78723292" | Completed             | "C019377721" | Go Sedan      | Nirman Vihar              | Malviya Nagar             | 15 | 25 | 470  | Cash        | 4.2 |
| 38 | 16-5-2024  | 13:04:31 | "CNR89118895" | Completed             | "C019520433" | Auto          | New Delhi Railway Station | Noida Film City           | 7  | 21 | 187  | UPI         | 4.3 |
| 39 | 2-10-2024  | 03:10:35 | "CNR7401001"  | Completed             | "C019958715" | Premier Sedan | Civil Lines Gurgaon       | Mandi House               | 7  | 30 | 186  | UPI         | 4.6 |
| 40 | 26-2-2024  | 15:43:16 | "CNR2631310"  | Completed             | "C014355670" | eBike         | Seelampur                 | Janakpuri                 | 10 | 36 | 70   | UPI         | 4.4 |
| 41 | 10-9-2024  | 13:02:42 | "CNR4218487"  | Cancelled by Customer | "C013037053" | Bike          | Noida Extension           | Udyog Vihar Phase 4       | 11 | 29 | 414  | Unknown     | 4.5 |
| 42 | 12-9-2024  | 09:14:57 | "CNR5333099"  | Cancelled by Driver   | "C015112050" | Auto          | Adarsh Nagar              | Civil Lines Gurgaon       | 8  | 29 | 414  | Unknown     | 4.5 |
| 43 | 24-1-2024  | 10:47:57 | "CNR5969954"  | Completed             | "C015558118" | Auto          | Vinobapuri                | Karkarduma                | 13 | 19 | 273  | Uber Wallet | 4.6 |
| 44 | 12-2-2024  | 19:42:39 | "CNR4130953"  | Incomplete            | "C013196782" | Premier Sedan | Panipat                   | Tagore Garden             | 6  | 11 | 966  | Debit Card  | 4.5 |
| 45 | 2-10-2024  | 19:17:00 | "CNR6693125"  | Cancelled by Driver   | "C017364449" | Auto          | Karol Bagh                | Tagore Garden             | 5  | 29 | 414  | Unknown     | 4.5 |
| 46 | 1-4-2024   | 21:14:19 | "CNR9572979"  | Completed             | "C013474992" | Go Sedan      | Sultanpur                 | Noida Extension           | 12 | 42 | 251  | Debit Card  | 3.6 |
| 47 | 18-9-2024  | 21:06:36 | "CNR6444364"  | Cancelled by Driver   | "C018262635" | Go Mini       | Moti Nagar                | Anand Vihar ISBT          | 7  | 29 | 414  | Unknown     | 4.5 |
| 48 | 2-8-2024   | 07:17:07 | "CNR4862806"  | Cancelled by Customer | "C017875150" | Auto          | Shastri Park              | Anand Vihar ISBT          | 11 | 29 | 414  | Unknown     | 4.5 |
| 49 | 30-9-2024  | 13:10:50 | "CNR4674790"  | Incomplete            | "C014785905" | Bike          | Dilshad Garden            | Nehru Place               | 2  | 30 | 453  | Debit Card  | 4.5 |
| 50 | 16-6-2024  | 00:44:47 | "CNR6866217"  | Completed             | "C017798304" | Go Sedan      | Dilshad Garden            | Nehru Place               | 8  | 42 | 246  | UPI         | 4.5 |
| 51 | 18-9-2024  | 09:33:55 | "CNR4932611"  | Completed             | "C018785963" | Auto          | Aya Nagar                 | Karkarduma                | 9  | 25 | 237  | UPI         | 3.3 |
| 52 | 4-5-2024   | 21:26:59 | "CNR1572483"  | Incomplete            | "C016059557" | Go Mini       | Rajiv Chowk               | Central Secretariat       | 10 | 22 | 1581 | Cash        | 4.5 |
| 53 | 3-6-2024   | 18:14:47 | "CNR6945858"  | Completed             | "C019079088" | Go Mini       | MG Road                   | Hauz Rani                 | 4  | 25 | 379  | Uber Wallet | 4.5 |
| 54 | 3-10-2024  | 22:20:34 | "CNR6584679"  | Completed             | "C014738537" | Go Mini       | Jasola                    | Palam Vihar               | 11 | 41 | 545  | Cash        | 4.3 |
| 55 | 25-5-2024  | 11:24:29 | "CNR6125653"  | Completed             | "C012808212" | Go Mini       | Kashmere Gate             | RK Puram                  | 3  | 35 | 479  | UPI         | 4.6 |
| 56 | 4-3-2024   | 17:46:46 | "CNR6092529"  | Cancelled by Driver   | "C015040634" | Go Sedan      | Ardee City                | Basai Dhankot             | 11 | 29 | 414  | Unknown     | 4.5 |
| 57 | 10-11-2024 | 05:13:57 | "CNR6339107"  | Completed             | "C017358763" | Go Mini       | Punjabi Bagh              | Madipur                   | 9  | 23 | 647  | Uber Wallet | 4.7 |
| 58 | 11-9-2024  | 10:36:44 | "CNR1349015"  | Cancelled by Driver   | "C019386278" | Auto          | Meerut                    | Badarpur                  | 4  | 29 | 414  | Unknown     | 4.5 |
| 59 | 11-4-2024  | 15:43:34 | "CNR9773309"  | No Driver Found       | "C019965847" | Uber XL       | Anand Vihar ISBT          | Dwarka Sector 21          | 8  | 29 | 414  | Unknown     | 4.5 |
| 60 | 1-6-2024   | 10:03:27 | "CNR2107544"  | Completed             | "C01266188"  | Premier Sedan | Mayur Vihar               | Palam Vihar               | 11 | 42 | 125  | Credit Card | 5.0 |
| 61 | 9-5-2024   | 10:44:30 | "CNR4171959"  | Cancelled by Driver   | "C018470723" | Premier Sedan | Lajpat Nagar              | Ramesh Nagar              | 6  | 29 | 414  | Unknown     | 4.5 |
| 62 | 30-7-2024  | 12:02:26 | "CNR5136307"  | Completed             | "C01816553"  | Premier Sedan | Tughlakabad               | Akshardham                | 12 | 22 | 375  | Cash        | 3.3 |
| 63 | 25-2-2024  | 16:23:19 | "CNR3635865"  | Completed             | "C016477355" | Auto          | Cyber Hub                 | Yamuna Bank               | 12 | 33 | 310  | Debit Card  | 3.8 |
| 64 | 2-3-2024   | 11:15:59 | "CNR2497989"  | Cancelled by Customer | "C015007066" | Auto          | Karkarduma                | IGI Airport               | 17 | 29 | 414  | Unknown     | 4.5 |
| 65 | 9-2-2024   | 04:59:03 | "CNR8946337"  | Cancelled by Driver   | "C016438326" | Go Sedan      | Shastri Nagar             | New Colony                | 5  | 29 | 414  | Unknown     | 4.5 |
| 66 | 27-8-2024  | 18:22:02 | "CNR3740694"  | Incomplete            | "C015885514" | Auto          | Aya Nagar                 | Green Park                | 2  | 10 | 370  | Cash        | 4.5 |
| 67 | 8-3-2024   | 09:28:38 | "CNR4660896"  | Cancelled by Driver   | "C015486947" | Premier Sedan | Dwarka Mor                | ITO                       | 6  | 29 | 414  | Unknown     | 4.5 |
| 68 | 20-3-2024  | 09:59:11 | "CNR2513330"  | Cancelled by Driver   | "C017805280" | Premier Sedan | Adarsh Nagar              | Noida Extension           | 7  | 29 | 414  | Unknown     | 4.5 |
| 69 | 25-1-2024  | 12:39:21 | "CNR2952424"  | Completed             | "C019413620" | eBike         | Vidhan Sabha              | Mehrauli                  | 12 | 18 | 291  | UPI         | 4.3 |
| 70 | 11-4-2024  | 11:14:35 | "CNR3096031"  | Completed             | "C013010926" | Auto          | Anand Vihar               | Saket                     | 6  | 24 | 114  | Uber Wallet | 4.5 |
| 71 | 17-6-2024  | 20:25:47 | "CNR4141159"  | Completed             | "C016810411" | Go Mini       | Uttam Nagar               | New Delhi Railway Station | 4  | 45 | 464  | Cash        | 4.2 |

# DATA PREPARATION STEPS

## Step 1: Import Data into Excel

- ✓ Loaded the raw Uber dataset into Excel.

## Step 2: Data Cleaning

- ✓ Removed duplicate columns
- ✓ Retained only relevant analytical fields
- ✓ Filled empty cells with NULL values
- ✓ Standardized data types

## Step 3: Feature Engineering

- ✓ Created calculated fields such as Hour, Completion Rate, Cancellation Rate, and Estimated Distance

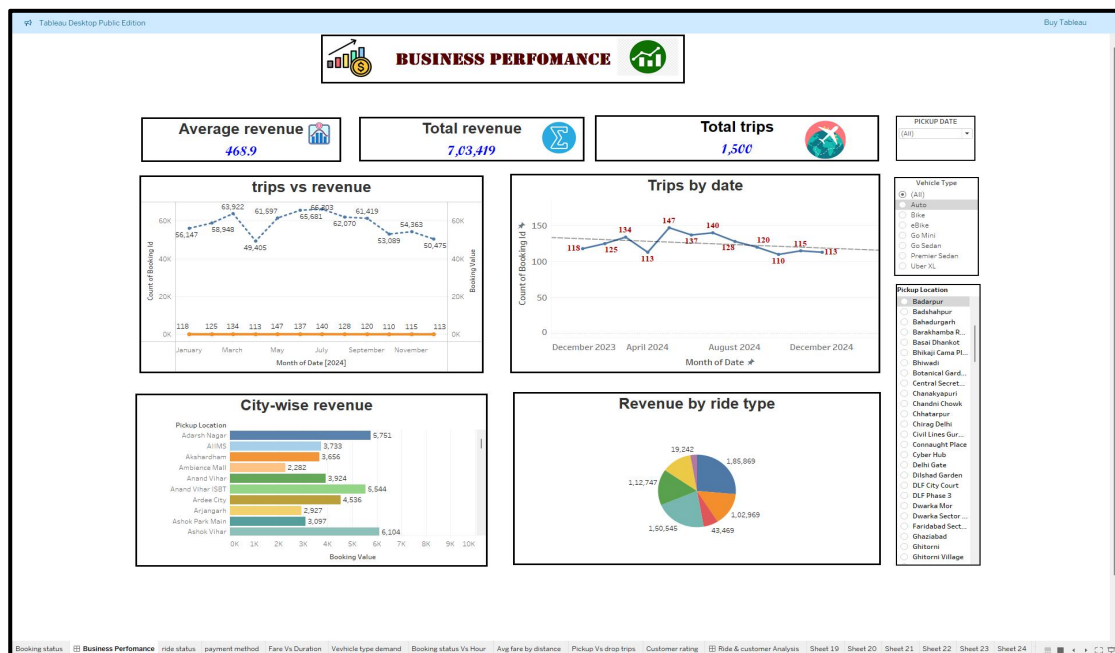
## Step 4: Import Data into Tableau

- ✓ Imported the cleaned dataset into Tableau Desktop

## Step 5: Visualization Development

- ✓ Designed KPIs, standard charts, and advanced visualizations

# DASHBOARD 1 – BUSINESS PERFORMANCE OVERVIEW



## KPIs

- **Total Trips** – COUNT(Booking Id)
- **Total Revenue** – SUM(Fare Amount)
- **Average Fare** – AVG(Fare Amount)

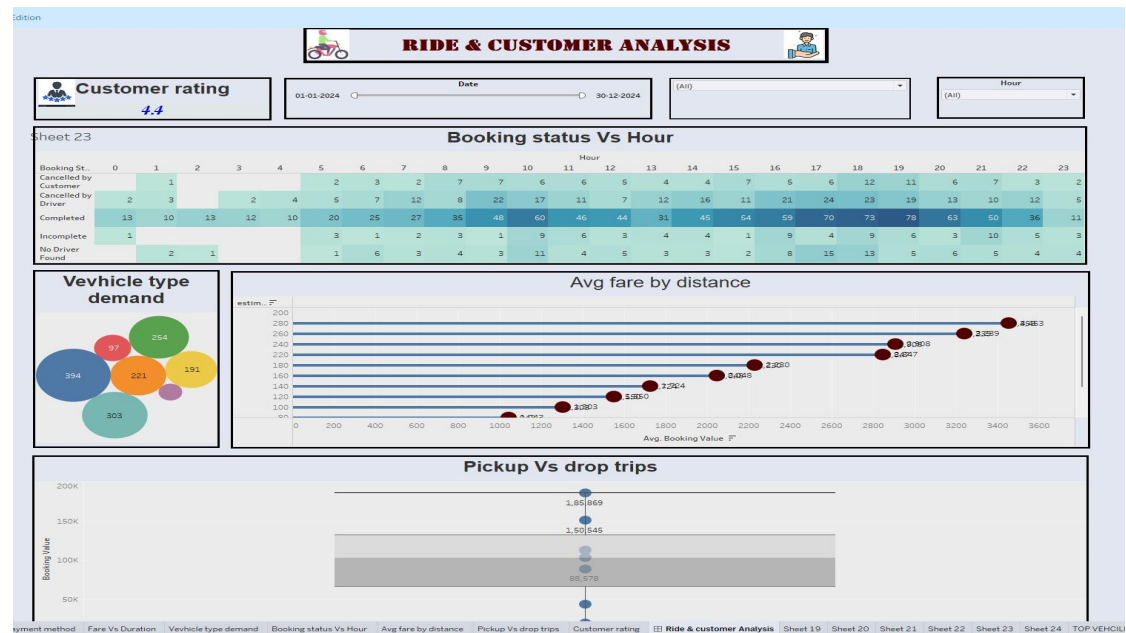
## VISUAL ANALYSIS

The dashboard provides a high-level overview of Uber's business performance. It includes time-series analysis of trips, city-wise revenue comparison, and vehicle-type revenue contribution. Advanced charts such as dual-axis trends and reference-line-based comparisons help identify growth patterns and top-performing segments.

## FILTERS USED

- Trip Date
- Pickup City
- Vehicle Type

# DASHBOARD 2 – RIDE & CUSTOMER ANALYSIS



## KPIs

- ✓ Average Customer Rating
- ✓ Average CTAT

## VISUAL ANALYSIS

This dashboard focuses on customer behavior and ride operations. It includes a 100% stacked bar for booking status, heatmap for booking status vs hour, bubble chart for vehicle demand, lollipop chart for average fare by estimated distance, and box plot for fare distribution by vehicle type. These advanced visuals help identify peak hours, cancellation trends, pricing variation, and service quality.

## FILTERS USED

- Trip Date
- Pickup City
- Vehicle Type
- Booking Status
- Hour

# **ANALYSIS SUMMARY & RECOMMENDATIONS**

## **Analysis summary**

- ✓ Peak booking hours show higher cancellation rates
- ✓ Certain vehicle types contribute disproportionately to demand
- ✓ Higher ratings are associated with better completion rates
- ✓ Fare distribution varies significantly across vehicle categories

## **Recommendations**

- ✓ Improve driver availability during peak hours
- ✓ Reduce customer cancellations through better ETA communication
- ✓ Optimize pricing strategies for high-demand vehicle types
- ✓ Focus on service quality to improve customer and driver ratings

## **CONCLUSION**

This Tableau project demonstrates how raw ride-hailing data can be transformed into insightful dashboards using effective data preparation, KPIs, and advanced visualizations. The dual-dashboard approach provides both strategic and operational perspectives, enabling better business decisions and enhanced customer experience.